Banking Management System

Software Documentation & User Guide

1. System Description

The Banking Management System is a desktop-based application developed using **Java Swing** and **MySQL**. It is designed to streamline essential banking operations, providing a secure and efficient interface for managing customer information and financial transactions.

Core Features:

- **Customer & Account Management:** Create, view, and update customer profiles and their associated bank accounts.
- **Transaction Processing:** Securely perform deposits and withdrawals with real-time balance updates.
- **Automated Receipting:** Instant pop-up transaction summaries with navigation options to history.
- **History Tracking:** A dedicated portal to view all historical transactions using a specialized Data Access Object (DAO) pattern.
- **Data Integrity:** Utilizes relational database constraints to ensure financial records remain consistent and accurate.

2. ERD Explanation (Database Architecture)

The system follows a relational database model to maintain data integrity and reduce redundancy. The Entity Relationship Diagram (ERD) consists of three primary entities:

Entity	Description	Relationship
Customer	Stores personal details (Name, Email, Phone).	One-to-Many with Account.

Entity	Description	Relationship
Account	Stores balance and account types (Savings/Current).	Many-to-One with Customer; One-to-Many with Transaction.
Transaction	Logs every financial movement (Type, Amount, Timestamp).	Many-to-One with Account.

Relationship Logic:

- **Customer → Account:** Linked via `customer_id`. This ensures every account is owned by a verified customer.
- **Account → Transaction:** Linked via `account_id`. This creates a permanent audit trail for every dollar moved within an account.
- **Constraints:** Foreign keys are set to `ON DELETE CASCADE` for accounts (if a customer is removed) and `RESTRICT` for transactions to maintain historical legal records.

3. How to Run the Program

Prerequisites:

- Java Development Kit (JDK) 8 or higher.
- MySQL Server (v8.0 recommended).
- MySQL Connector/J Library (JDBC Driver).
- NetBeans IDE or any Java-compatible IDE.

Setup Steps:

Step 1: Database Setup

Open MySQL Workbench and execute your SQL script to create `bankingsystemdb`. Ensure the tables `customer`, `account`, and `transaction` are created.

Step 2: Database Connection

Open `DBConnection.java` in the `bankingsystem.dao` package and update your credentials:

```
String url = "jdbc:mysql://localhost:3306/bankingsystemdb";
String user = "your_username";
String password = "your_password";
```

Step 3: Compile & Build

In NetBeans, right-click the project folder and select **Clean and Build**. This ensures all DAOs and Frames are properly compiled.

Step 4: Launch

Right-click `MainDasboard.java` and select **Run File**. The application will launch at the center of your screen.